



PROJECT REPORT

CAMPUS NAVIGATOR

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Abstract

The Campus Navigator System is a web-based website developed to help students, visitors, and staff easily navigate a university campus. The website provides detailed information about campus departments and locations through QR code scanning. Users can scan QR codes placed at different locations to instantly access relevant information without manual searching. The system is built using HTML, CSS, and JavaScript for the frontend, with Node.js used for backend functionality. The website aims to improve campus accessibility, assist new users in finding locations efficiently, and provide a simple, user-friendly navigation solution through a centralized web platform.

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1. Project Overview

Navigating a large university campus can be difficult for new students, visitors, and even staff, especially when searching for specific departments or facilities. The Campus Navigator System was developed as a web-based solution to address this challenge by providing quick and easy access to campus location information. The website allows users to scan QR codes placed at different campus locations to instantly view department details and navigation-related information. Built using HTML, CSS, and JavaScript for the frontend, with Node.js handling backend operations, the system ensures efficient data handling and smooth user interaction. The project aims to simplify campus navigation, reduce confusion, and enhance accessibility through a centralized and user-friendly website.

2. Objectives

- To develop a user-friendly web interface for accessing campus and department information.
- To integrate QR code scanning functionality for quick location-based data retrieval.
- To provide accurate and structured information about campus departments and facilities.
- To implement backend support using Node.js for efficient data processing.
- To improve navigation efficiency and overall user experience within the university campus.

3. Technology Stack:

Component	Technology Used	Purpose
i.Frontend	HTML, CSS, JavaScript	Used to design the structure, layout, and interactivity of the website
ii. Backend	Node.js	Handles server-side logic and request processing
iv. Development Tools	Visual Studio Code	Used for writing and managing source code
v. Testing Platform	Web Browsers (Google Chrome/Microsoft Edge)	Used for testing and debugging the website

4. Website Architecture

The Campus Navigator Website is modularly structured to ensure maintainability, scalability, and ease of navigation. Its main components include:

- **Pages / Screens:** Home, Department Details, Navigation Guide.
- **Components:** DepartmentCard to display information about each department or facility.
- **QR Code Module:** Handles QR code scanning and retrieval of location-specific information.
- **Backend Module:** Node.js server manages requests, fetches data, and serves responses.
- **Utilities:** Functions for data handling, formatting, and efficient information display.
- **Navigation:** Easy-to-use web navigation links and QR-driven navigation for seamless movement across sections.

Data persistence is managed through structured storage on the backend, ensuring accurate retrieval of campus information whenever required.

5. Key Features

- **Department Information:** Displays details of all campus departments, including name, location, and additional information.
- **QR Code Scanning:** Allows users to scan QR codes placed across the campus to quickly access relevant information.
- **Dynamic Navigation:** Provides direct links and guidance to various campus locations based on scanned QR codes.
- **User-Friendly Interface:** Easy-to-navigate website with clear layouts and interactive components.
- **Efficient Data Management:** Ensures department data is consistently updated and accessible through the backend server.

6. Code Highlight and Explanation:

Splash Screen Snippet:

```
setTimeout(function() {  
    window.location.href = 'index.html';  
}, 5000);
```

Explanation:

Displays loading screen for 5 seconds then automatically redirects to main application.

Department Filtering Snippet:

```
function performSearch() {  
    const query = searchInput.value.toLowerCase().trim();  
    let filteredDepts = departments;  
  
    if (query) {  
        filteredDepts = filteredDepts.filter(dept =>  
            dept.name.toLowerCase().includes(query) ||  
            dept.location.toLowerCase().includes(query)  
        );  
    }  
  
    if (activeFilter !== 'all') {  
        filteredDepts = filteredDepts.filter(dept =>  
            dept.type === activeFilter  
        );  
    }  
  
    displayResults(filteredDepts);  
}
```

Explanation: Combines text search with category filters to dynamically display relevant campus departments.

Feedback System with Star Rating

javascript

```
stars.forEach(star => {  
  star.addEventListener('click', () => {  
    const rating = star.dataset.rating;  
    ratingInput.value = rating;  
    stars.forEach((s, index) => {  
      s.classList.toggle('active', index < rating);  
    });  
  });  
});
```

Explanation: Interactive star rating system where clicking stars highlights them and stores rating value.

Report Submission API (server.js)Snippet:

```
app.post('/api/report-issue', (req, res) => {  
  const { name, contact, issueType, details, location } = req.body;  
  const reportsPath = path.join(__dirname, 'backend', 'data', 'reports.db');  
  // Save to JSON file  
  fs.writeFileSync(reportsPath, JSON.stringify(reportsData, null, 2));  
  res.json({ success: true, message: "Issue reported successfully!" });  
});
```

Explanation: Express.js endpoint that receives report form data and saves to JSON database file.

1. QR Scanner - Camera Start Function

javascript

```
function startCamera() {  
  navigator.mediaDevices.getUserMedia({
```



```

        video: { facingMode: "environment" }
    }).then(s => {
        stream = s;
        video.srcObject = stream;
        video.play();
        scanning = true;
        requestAnimationFrame(scanFromCamera);
    });
}

```

Explanation: Requests camera access, starts video stream for QR scanning, and begins continuous scanning loop.

2. QR Detection Handler (scanner.html)

```

javascript
function onQRDetected(qrCodeID) {
    console.log("QR Detected:", qrCodeID);
    stopCamera();
    fetch(`/get-guide/${qrCodeID}`)
        .then(res => res.json())
        .then(data => {
            if (data.success) {
                window.location.href = data.url;
            }
        });
}

```

Explanation: When QR code is detected, stops camera, fetches corresponding video guide from backend, and redirects.

3. QR Code Video Mapping API (server.js)

javascript

```
app.get('/get-guide/:id', (req, res) => {  
  const qrId = req.params.id;  
  const campusData = JSON.parse(fs.readFileSync('backend/data/campus.db'));  
  const guide = campusData.find(item => item.qrCode === qrId);  
  if (guide) {  
    res.json({ success: true, url: guide.url });  
  }  
});
```

Explanation: Looks up QR code in database and returns corresponding video URL for navigation.

Video Popup Player (index.html)

```
function playVideoPopup(videoPath, title) {  
  currentVideoSrc = videoPath;  
  document.getElementById('videoSource').src = videoPath;  
  document.getElementById('videoPlayer').load();  
  document.getElementById('videoPopup').style.display = 'flex';  
}
```

Explanation: Opens modal popup with video player and loads specified video file for campus location.

ScreenShots:



Welcome to FJWU Campus Navigator

Using QR codes Explore your Campus with Ease

Figure 1: Welcome Screen

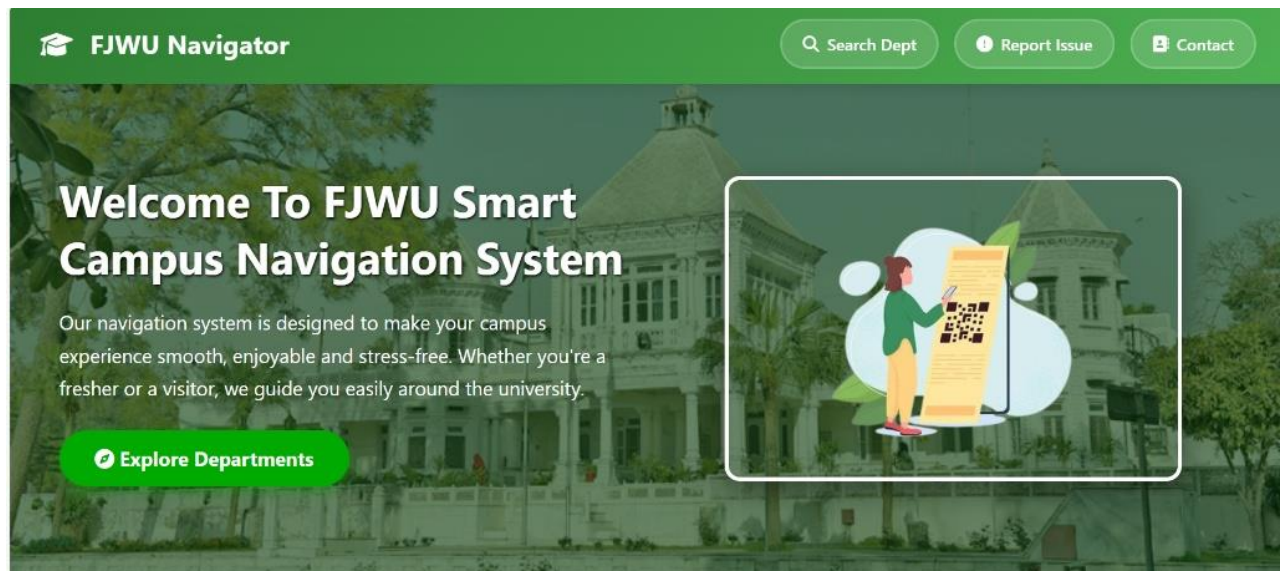


Figure 2: Landing page

The main landing page providing navigation links and a direct 'Explore Departments' action button.

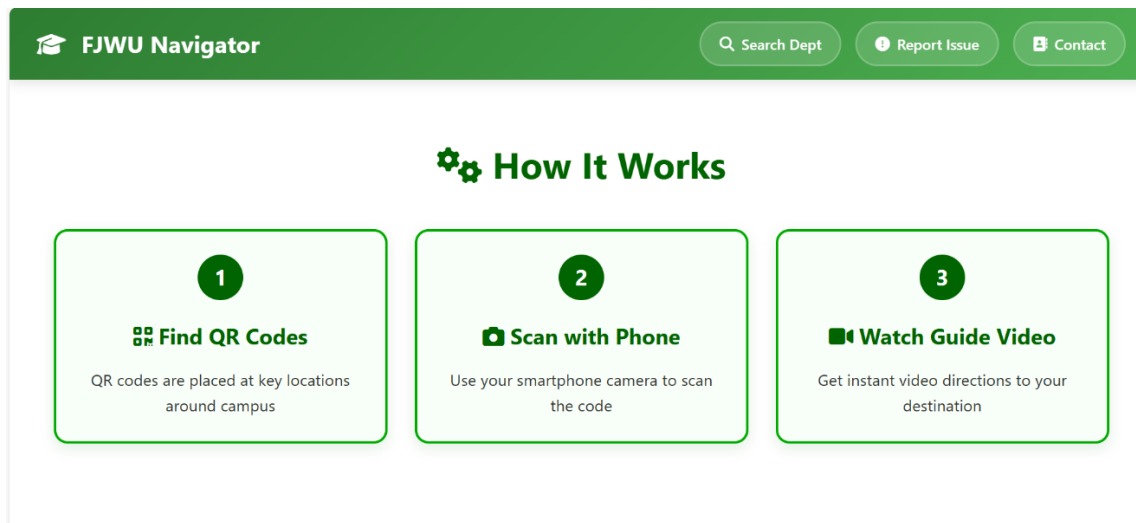


Figure 3: FJWU Navigator instructional UI layout.

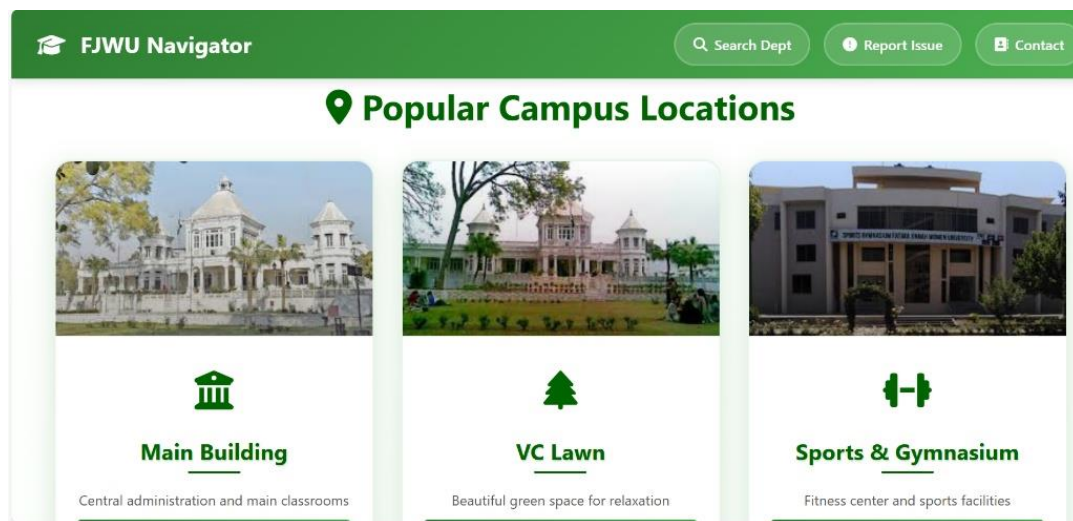


Figure 4: Popular Campus Locations UI layout

FJWU Navigator

Welcome To FJWU Campus Navigation

Our navigation system is designed to make your experience smooth, enjoyable and stress-free. Whether you are a fresher or a visitor, we guide you easily and quickly.

[Explore Departments](#)

Your Name *

Email/Phone *

Issue Type *

Select issue type

Details *

Please describe the issue in detail...

Location

Where did this happen?

[Submit Report](#) [Cancel](#)

Report Issue **Contact**

Figure 5: User interface for the FJWU Navigator reporting system

[Back to Home](#) **Department Search**
Find any campus location

Search Campus Departments

Type department name, building, or use filters

Example: Computer Science, Library, Administration...

[All](#) [Academic](#) [Student Services](#) [Feedback](#)

[Search Department](#)

Figure 6: Department search and filter UI

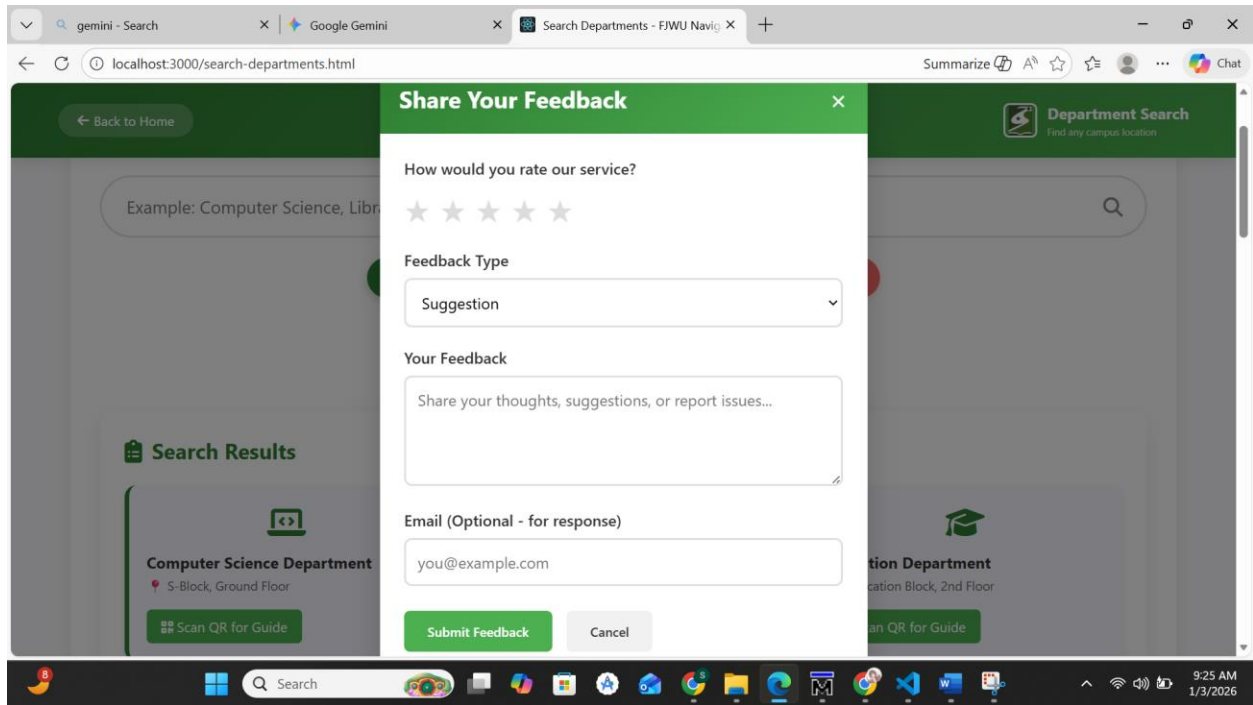


Figure 7: User feedback and rating modal

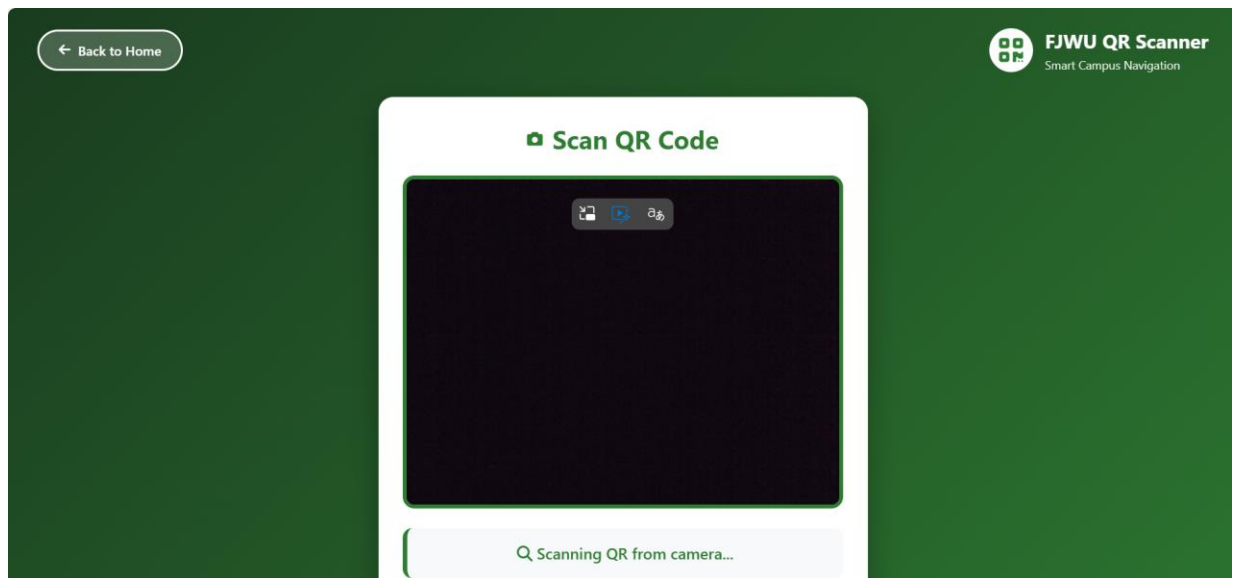


Figure 8: Integrated QR code scanner interface

7. Conclusion

The FJWU Campus Navigator website successfully addresses the challenges of campus navigation by providing an efficient, user-friendly, and QR-based solution. The system enables students and visitors to quickly access department information, locations, and multimedia guidance through an interactive web interface. By integrating QR code scanning, structured data handling, and responsive design, the project enhances accessibility and reduces confusion within the campus environment. Overall, the Campus Navigator demonstrates the effective use of modern web technologies to solve real-world problems